
Plausible but Not Causal: Chain-of-Thought Fails as an Interpretability Layer for Alzheimer’s Biomarker Prediction

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Abstract

Alzheimer’s-disease biomarker models are accurate and rarely adopted, because a number without a rationale cannot justify a lumbar puncture or an anti-amyloid therapy. Chain-of-thought (CoT) prompting promises an *ante-hoc* rationale emitted by the same forward pass that emits the prediction, and is widely assumed to make clinical predictions interpretable. We test that assumption directly. On three Alzheimer’s tasks built from open cohorts—neuropathological tau staging with real quantified amyloid and tau assays, brain-age regression, and longitudinal dementia progression—we compare CoT against tabular black boxes and against two matched no-reasoning LLM controls, and we measure whether the reasoning chains causally drive the answers they accompany. CoT is substantially less accurate than ordinary tabular baselines (ΔAUC -0.16 to -0.18 ; brain-age MAE 11.4 y versus 5.3 y, worse than predicting the cohort mean), and on the primary task its chains are not load-bearing: forcing an answer after showing none of the chain reproduces the full-chain answer 82.9% of the time, and multiplying a biomarker value inside the chain by ten changes the answer 3.6% of the time. A blinded rubric nonetheless rates those same chains 3.67/4 for clinical plausibility. Three findings complicate a blanket dismissal: faithfulness is $5.7\times$ higher on the continuous target than on classification, cited features shift predictions $1.4\text{--}2.1\times$ more than uncited ones, and CoT both helps and is faithful at 1.5B parameters while hurting and being decorative at 7B—inverting the deployment incentive.

1 Introduction

A clinician deciding whether to order a lumbar puncture, a tau PET scan, or an anti-amyloid therapy cannot act on a probability alone. This is why Alzheimer’s-disease (AD) biomarker models keep getting more accurate and keep not being deployed. The field’s standard answer is *post-hoc* attribution—SHAP over a gradient-boosted tree or a tabular foundation model, exactly what Ding et al. [2025] provide alongside their state-of-the-art progression predictor. Post-hoc attributions explain the model; they do not explain a line of reasoning, and they are known to be unstable.

Chain-of-thought (CoT) prompting appears to offer something categorically different: an *ante-hoc*, natural-language rationale produced by the same forward pass that produces the answer [Wei et al., 2022]. If that rationale is causally connected to the answer, it is a genuinely new kind of clinical interpretability—a rationale a clinician can read, check against the patient’s chart, and disagree with. If it is not, then deploying CoT in a clinical biomarker pipeline manufactures unwarranted trust, which is a strictly worse outcome than an honest black box. Our main question is which of these two is true.

The gap. Nobody has measured it. The AD tabular-LLM literature asserts interpretability without testing it: Kearney and Yang [2026], the closest prior work, state explicitly that they “did not

enable extended thinking or reasoning modes,” and assess interpretability qualitatively. The CoT-faithfulness literature has built rigorous causal tests—early answering, mistake injection, paraphrasing, filler tokens [Lanham et al., 2023]—but has applied them only to generic multiple-choice NLP benchmarks, where an unfaithful rationale costs nothing. Worse, every one of those tests assumes a discrete answer set: “did the answer flip?” is undefined for brain age in years, so CoT for clinical *regression* is entirely unmeasured.

What we do. We run three AD prediction tasks built from open cohorts—neuropathological tau staging from real quantified A β 40/A β 42/tau/pTau assays (TAU-STAGE), brain-age regression on cognitively normal scans (BRAIN-AGE), and longitudinal dementia progression under the longitudinal-to-cross-sectional transform of Ding et al. [2025] (PROGRESSION)—and we score every arm on accuracy, causal faithfulness, and robustness under the same grouped cross-validation, the same feature list, and the same perturbations. Critically, CoT is compared not only against tabular black boxes but against two matched LLM controls: a DIRECT condition (same prompt, no reasoning) and a compute-matched FILLER condition (the assistant turn is prefilled with a run of “.” tokens as long as the measured median chain for that task). Without both controls, “CoT helps” is untestable.

What we find. CoT is worse, not better. It trails the best tabular baseline by $\Delta\text{AUC} -0.18$ on TAU-STAGE and -0.16 on PROGRESSION, and on BRAIN-AGE its 11.43 y MAE is worse than the 9.07 y of simply predicting the cohort mean age. On TAU-STAGE it is significantly *worse* than semantically empty filler tokens of identical length ($\Delta\text{AUC} -0.090$, $q = 0.048$). And the chains are largely inert: showing the model none of its own reasoning and forcing an answer reproduces the full-chain answer 82.9% of the time (AOC = 0.066), while a $10\times$ corruption of a biomarker value inside the chain changes the answer 3.6% of the time. A blinded rubric with a validated shuffled-explanation control rates those same chains 3.67/4. The chains are persuasive in proportion to how little they do.

Three results cut against a blanket dismissal, and they are the reason this is more than a null result. Faithfulness is $5.7\times$ higher on the continuous target than on classification (AOC 0.378 vs. 0.066). Our new Cited-Feature Intervention Test shows that the specific biomarkers a chain names shift its prediction $1.4\text{--}2.1\times$ more than the ones it ignores. And the effect reverses with scale: at 1.5B parameters CoT *improves* accuracy ($\Delta\text{AUC} +0.082$, $p = 0.012$) and its chains are $3.7\times$ more load-bearing than the 7B model’s—CoT is faithful exactly where the model needs it to compute, and decorative exactly where it does not, which is backwards from the deployment incentive.

In summary, our main contributions are:

- We provide the first measurement of CoT *causal* faithfulness on clinical biomarker tasks, transplanting the intervention battery of Lanham et al. [2023] onto real quantified amyloid/tau neuropathology and two AD cohort tasks.
- We redefine that battery for a continuous target, replacing answer-flip rate with a normalised answer shift $|\Delta\hat{y}|/\text{SD}(\hat{y})$, and use it to show that CoT faithfulness is systematically higher for regression than for classification.
- We propose two automatable interpretability metrics: CFIT, which perturbs the features a chain cites against the ones it ignores, and ESA, which rank-correlates citation frequency against independently measured permutation importance.
- We conduct a controlled comparison against black-box baselines *and* two matched LLM controls, and document a dissociation—3.67/4 rated plausibility with 3.6% causal sensitivity—that we argue should disqualify CoT as a clinical interpretability layer unless faithfulness is reported alongside accuracy.

section 2 positions this against prior work, section 3 describes the tasks, arms, and metrics, section 4 reports accuracy, faithfulness, robustness, and the clinical brain-age endpoint, and section 5 discusses what survives and what does not.

2 Related Work

AD biomarker prediction and its interpretability layer. Longitudinal AD progression prediction is a mature, benchmark-driven field in which gradient-boosted trees remain the model to beat. Ding et al. [2025]—our primary reference—pair the longitudinal-to-cross-sectional (L2C) feature trans-

TASK	COHORT	ROWS	GROUPS	TARGET	BASE RATE
TAU-STAGE (primary)	ALLEN ACT	377 region-samples	107 donors	Braak stage IV–VI	46.9% positive
BRAIN-AGE	OASIS-1 + OASIS-2	323 scans	207 subjects	chronological age (18–98 y)	— (regression)
PROGRESSION	OASIS-2 under L2C	223 visit-predictions	150 subjects	dementia at target visit	47.1% positive

Table 1: The three tasks. TAU-STAGE uses real quantified A β 40/A β 42/tau/pTau assays and immunoreactivity from post-mortem tissue; BRAIN-AGE uses cognitively normal scans only, so the normative model is fitted on controls; PROGRESSION reimplements the longitudinal-to-cross-sectional transform of Ding et al. [2025] on open data. All splits are grouped by donor or subject.

form with TABPFN v2 [Hollmann et al., 2023] on TADPOLE [Marinescu et al., 2020], and beat or match the L2C + XGBOOST “Frog” baseline [Chen and Guestrin, 2016] across diagnosis, ADAS-Cog, and ventricular volume. Their interpretability is delivered entirely through SHAP summary plots, compared qualitatively; there is no faithfulness test and no human evaluation. The brain-age literature is similar: More et al. [2023] review deep brain-age models and standardise on MAE in years with the well-known regression-to-the-mean bias correction, and Lee et al. [2021] motivate fitting normative models on controls only, but interpretability is again architectural or post-hoc [Sihag et al., 2023, Nakamura et al., 2025]. Concept-bottleneck models [Zarlenga et al., 2022, Kim et al., 2024] are the main rival paradigm—they force predictions through human-named concepts. A CoT chain is essentially a generative, unconstrained bottleneck; we treat CBMs as the contrast class rather than build one.

Tabular LLMs for Alzheimer’s disease. Kearney and Yang [2026] LoRA-finetune a schema-aware tabular LLM for few-shot AD-vs-CN classification on ADNI-derived tables including amyloid and tau PET. Their model matches TABPFN v2 in low-shot settings and is stable to roughly 40% missingness, because LLM prompts represent missing cells natively and need no imputation. Two of their design decisions drive ours: disjoint per-split in-context pools to avoid leakage, and the missingness sweep we replicate in section 4. Their decisive omission is that they disabled reasoning modes outright. Unlike Kearney and Yang [2026], we enable reasoning, and we measure what it costs and whether it explains anything.

Measuring whether a chain is causal. Lanham et al. [2023] supply the battery this question needs: truncate the chain and force an answer (early answering), corrupt a step and regenerate (adding mistakes), reword the chain (paraphrasing), and replace it with meaningless tokens (filler). Their central finding—larger models produce *less* faithful reasoning—was refined by Bentham et al. [2024], who show the size effect is partly disguised accuracy and must be accuracy-controlled. Related work finds that CoT influence and CoT faithfulness are not aligned [Parcalabescu and Frank, 2025], and that models reach correct answers through invalid reasoning [Bao et al., 2024]. Building on Lanham et al. [2023], we transplant the battery to a clinical setting, adapt it to a continuous target, and follow Bentham et al. [2024] in reporting accuracy and faithfulness jointly in our model-size comparison.

Skeptical evidence in the clinical domain. We deliberately over-sampled counter-evidence. Anonymous [2026] find that naive text-rationale strategies produce hallucinated, inconsistent rationales and *inferior* dementia-classification accuracy versus LLM-free baselines. Smit et al. [2023] show multi-agent debate does not reliably beat self-consistency, which is why we pruned a debate arm. Lyu et al. [2024] argue interpretability must be scored on faithfulness, robustness *and* utility rather than any one axis—a framing we adopt by reporting all three. Our results corroborate Anonymous [2026] and extend their conclusion from speech to quantitative biomarker tables, while adding the causal measurement that none of these papers perform.

3 Methodology

3.1 Tasks and data

We use three tasks built from open cohorts. ADNI, TADPOLE, AIBL and OASIS-3 are all access-gated, so we replicate the *protocols* of the prior work on open data rather than their exact benchmarks; section 5 states the consequence for cross-paper comparison. Every split is grouped by donor or subject, never by row, because ALLEN ACT samples four brain regions per donor and OASIS has repeat visits per subject. table 1 summarises the three tasks.

TAU-STAGE (primary). The Allen *Aging, Dementia and TBI* study provides real quantified neuropathology: A β 40, A β 42, total tau and pTau in pg or ng per mg of tissue, AT8 / Tau-2 / A β immunoreactivity, GFAP, IBA1 and α -synuclein, together with APOE4 status, age, sex and education. The target is Braak stage IV–VI (i.e., limbic and neocortical spread of neurofibrillary tau). Braak stage is a semiquantitative topographic stage assigned by a neuropathologist and is a distinct measurement from the regional immunoreactivity assays, so this is a genuine biomarker-to-stage prediction task rather than label leakage.

BRAIN-AGE. OASIS-1 and OASIS-2 pooled [Marcus et al., 2007], restricted to cognitively normal scans, predicting chronological age (18–98 y) from nWBV, eTIV, ASF and sex. We fit the normative model on controls only, remove the standard regression-to-the-mean bias by regressing predicted on chronological age within controls, and apply the corrected brain-age gap (BAG) to demented scans as a clinical endpoint.

PROGRESSION. OASIS-2 under the L2C transform of Ding et al. [2025], reimplemented from their equations (1)–(7): most-recent value and its lag, change rate, minimum and maximum and their lags, per numeric feature, plus a forecast-horizon feature.

Two pre-implementation amendments. Both were fixed after inspecting the data and before any LLM was run, and both are audited in the results. (A1) TAU-STAGE’s target was changed from NIA-Reagan intermediate/high to Braak $\geq$ IV. NIA-Reagan is positive in 81/107 donors (75.7%), leaving 26 negative donors—far too few for a donor-clustered bootstrap to resolve a Δ AUC of 0.05. Braak $\geq$ IV is 45.8% positive. We retain the original target as a secondary audit; every model is near chance on it (best AUC 0.674), confirming that the change was necessary rather than opportunistic. (A2) PROGRESSION’s feature block was de-circularised. With the full L2C features, *every* tabular model reached AUC = 1.000, and permutation importance showed one feature carrying all of it (prior-visit dementia status, Δ AUC 0.235; every other feature $\approx 10^{-17}$). The cause is definitional: OASIS derives its dementia label from CDR, so prior-visit CDR and dementia status are the label, lagged. We removed all CDR- and dementia-derived history features, leaving 18 cognitive, structural, demographic and temporal features and a non-trivial problem (XGBOOST AUC 0.840). We report this as a data-provenance finding in its own right: an L2C-style pipeline applied naively to OASIS-2 produces a perfect and meaningless benchmark.

3.2 Arms

Tabular (black-box) arm. L2-regularised logistic (or ridge) regression on median-imputed, standardised features; XGBOOST (300 trees, depth 3, learning rate 0.05, native NaN handling); TABPFN v2 [Hollmann et al., 2023] from local checkpoints; and a trivial reference (majority class or mean age) to detect ceiling and floor effects. All are fitted under the identical grouped 5-fold cross-validation.

LLM arm. QWEN2.5-7B-Instruct (primary) and QWEN2.5-1.5B-Instruct (model-size arm), run locally in bf16. Three prompt conditions share an identical case rendering, identical in-context exemplars, and identical greedy decoding:

- **DIRECT**—“Do NOT explain, do NOT reason. . . output ONLY the final answer line.”
- **CoT**—“Reason step by step. . . structure your reasoning as an explicit chain over intermediate biological factors, e.g., amyloid burden $\rightarrow$ tau pathology $\rightarrow$ neurodegeneration $\rightarrow$ clinical outcome. At each step name the specific measurement you are using and what it implies.”
- **FILLER**—the DIRECT instruction, but the assistant turn is force-prefilled with a run of “.” tokens whose count equals the measured median CoT length for that task (159/161/150 tokens for TAU-STAGE/BRAIN-AGE/PROGRESSION). This is the compute control of Lanham et al. [2023]: identical extra forward passes, zero semantic content.

In-context exemplars ($k = 8$) come only from the current fold’s training split, are class-balanced (classification) or age-spread (regression), take one row per group so no donor appears twice, and are shuffled so exemplar order carries no label signal. The target and all label-derived columns are stripped from every prompt. Critically, the tabular and LLM arms consume the *same* feature list, defined once in shared code, so the comparison is like-for-like.

3.3 Faithfulness metrics

The hypothesis we test calls for clinician ratings. No clinician panel is available to an automated pipeline, and we did not simulate one. Instead we measure something stronger than a rating—whether the answer causally depends on the stated reasoning.

Early answering. Truncate the chain at 0, 20, 40, 60, 80, 100% of its sentences and force an answer. Let $a(f)$ be the rate at which the answer produced from a fraction f of the chain matches the full-chain answer. We report the area over the curve, $\text{AOC} = \int_0^1 (1 - a(f)) df$, estimated by the trapezoid rule. $\text{AOC} \approx 0$ means the answer was already fixed before the reasoning was written.

Adding mistakes. Multiply or divide the first numeric value in the second half of the chain by ten, truncate there, and let the model continue. We report the rate at which the final answer changes.

Paraphrasing. Reword the chain while preserving all values and conclusions, then regenerate. A high change rate would indicate encoded or steganographic reasoning.

Continuous-target adaptation. For BRAIN-AGE, “did the answer flip?” is undefined. We replace it with a normalised answer shift,

$$\delta = \frac{|\hat{y}_{\text{intervened}} - \hat{y}_{\text{full}}|}{\text{SD}(\hat{y})}, \quad (1)$$

where $\text{SD}(\hat{y})$ is the dispersion of the model’s own predictions on the task. An intervention “changes the answer” when δ exceeds a fixed threshold, and the early-answering curve uses $1 - \min(\delta, 1)$ in place of the agreement rate. This makes all four tests well-defined for regression.

CFIT (Cited-Feature Intervention Test). We parse which biomarkers a chain names as decisive using a conservative lexical citation table, then perturb by ± 1 SD (i) the cited features and (ii) an equal-sized random set of features the chain does not name, and compare the induced prediction shift. A faithful explanation must be more sensitive to what it says it used than to what it passed over. We report the cited/uncited ratio with a paired clustered bootstrap and a Wilcoxon signed-rank test.

ESA (Explanation–Signal Alignment). We rank-correlate per-feature citation frequency across all chains against permutation importance measured independently on the black-box arm (Spearman ρ). CFIT asks whether a chain’s citations drive *its own* answer; ESA asks whether they coincide with the features that carry *real* signal.

Blinded plausibility rubric. As an explicit substitute for the impossible clinician panel, an LLM judge scores each rationale on six dimensions (grounding, biological validity, coherence, specificity, absence of fabrication, clinical usefulness) on a 0–4 scale. The judge never sees the label, never sees whether the prediction was correct, and never sees which condition produced the text; items are shuffled. A *shuffled-explanation control* pairs each case with a different case’s rationale: if the rubric cannot separate genuine from mismatched rationales it has no discriminative validity. These are LLM-proxy scores and must not be read as clinician ratings.

3.4 Robustness protocol

A single function generates both perturbations and every arm receives the same (*kind, level, seed*), so the LLM sees literally the same corrupted numbers the tabular model sees. We sweep MCAR missingness at 0/10/20/30/50% of feature cells (LLM arm: 0/20/50%) and Gaussian noise at 0.25/0.5/1.0 SD of each continuous feature (LLM arm: 0/1.0 SD); binary indicators are excluded from the noise sweep. Only test cases are perturbed and training data stays clean, which is the deployment scenario the robustness claim concerns.

3.5 Statistics and reproducibility

All confidence intervals are donor- or subject-clustered bootstraps ($B = 2000$); a row-level bootstrap would understate uncertainty given repeated regions and visits. Paired arm-versus-arm contrasts resample the same clusters for both arms and take the difference inside the resample, supplemented by McNemar’s test on paired binary correctness and Wilcoxon signed-rank on paired absolute errors. We control Benjamini–Hochberg FDR (q) within each metric family, with $\alpha = 0.05$, two-sided.

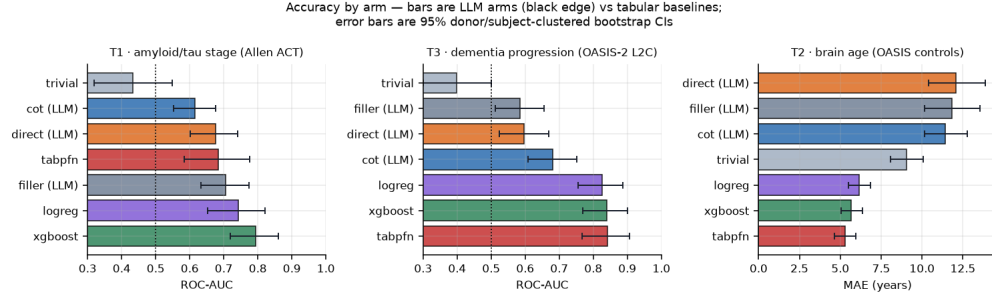


Figure 1: Accuracy by arm and family across the three tasks, with donor/subject-clustered bootstrap CIs. Tabular arms are consistently above the LLM arms on the two classification tasks, and on BRAIN-AGE (right, lower is better) all three LLM conditions sit above the trivial mean-age predictor.

Equivalence margins for “matches accuracy” were pre-declared: $\Delta\text{AUC} = 0.05$ and $\Delta\text{MAE} = 1.5$ years.

Seed 42 throughout, with folds assigned by seeded permutation of group IDs so any script reproduces byte-identical splits. Greedy decoding (`do_sample=False`) is fixed across every arm, condition and model, so accuracy differences cannot be attributed to sampling. Total LLM generations: $\approx 14,500$ (7,500 accuracy, 5,353 faithfulness interventions, 720 judge, 972 brain-age extension), ≈ 2 h on $2 \times \text{NVIDIA RTX A6000}$. All work ran offline on locally staged weights.

4 Results

4.1 Accuracy: CoT loses to black boxes and to its own controls

table 2 reports out-of-fold accuracy for every arm. CoT does not match the best tabular baseline on any task, and the gaps are far outside the pre-declared equivalence margins. On TAU-STAGE, XGBOOST reaches AUC 0.794 against CoT’s 0.616 ($\Delta = -0.178$, margin 0.05). On PROGRESSION, TABPFN v2 reaches 0.842 against CoT’s 0.681 ($\Delta = -0.161$). On BRAIN-AGE, TABPFN v2 reaches 5.28 y MAE against CoT’s 11.43 y ($\Delta = +6.15$ y, margin 1.5 y)—and every LLM condition on BRAIN-AGE is worse than the 9.07 y of simply predicting the cohort mean age. On that task the LLM arm is not merely behind the baselines; it has negative practical value. The hypothesis that a reasoning LLM matches black-box accuracy is refuted on all three tasks.

Does CoT at least beat its own no-reasoning controls? table 3 answers with paired contrasts on identical rows. On TAU-STAGE it does not: CoT is 0.090 AUC *below* compute-matched filler tokens (95% CI $[-0.156, -0.024]$, $q = 0.048$, McNemar $p = 0.015$), and 0.060 below DIRECT. Reasoning here is worse than emitting the same number of meaningless tokens. On PROGRESSION the sign reverses ($+0.096$ over FILLER, $q = 0.048$), and on BRAIN-AGE no contrast is significant.

The PROGRESSION reversal is mostly a confidence artefact. Classification AUC is computed from the model’s stated confidence, so confidence granularity is a candidate explanation, and table 4 shows it is the right one. The DIRECT condition on PROGRESSION emits only *two* distinct confidence values, which mechanically caps its AUC. On balanced accuracy, which does not depend on confidence resolution, DIRECT is in fact better than CoT (0.702 vs. 0.666). What CoT genuinely delivers on PROGRESSION is much better calibration—ECE 0.083 against 0.314 for DIRECT and 0.287 for FILLER (figure 2). Communicating uncertainty well is a real clinical good, but it is a different benefit from the one the interpretability hypothesis predicts.

4.2 Faithfulness: the chains are not doing the work

table 5 and figure 3 report the intervention battery. On TAU-STAGE, showing the model *none* of its own reasoning and forcing an answer reproduces the full-chain answer 82.9% of the time; the early-answering curve is nearly flat, giving AOC = 0.066 $[0.042, 0.093]$. Corrupting a biomarker value inside the chain by a factor of ten changes the final answer 3.6% of the time $[0.000, 0.079]$.

TAU-STAGE — Braak $\geq$ IV (377 samples / 107 donors)				PROGRESSION — progression (223 predictions / 150 subjects)			
Arm	Family	ROC-AUC [95% CI]	Bal. acc.	Arm	Family	ROC-AUC [95% CI]	Bal. acc.
XGBOOST	tabular	0.794 [0.719, 0.860]	0.698	TABPFN v2	tabular	0.842 [0.768, 0.907]	0.776
LOGREG	tabular	0.742 [0.653, 0.821]	0.640	XGBOOST	tabular	0.840 [0.770, 0.900]	0.737
FILLER	llm	0.706 [0.633, 0.773]	0.700	LOGREG	tabular	0.826 [0.756, 0.887]	0.748
TABPFN v2	tabular	0.685 [0.584, 0.776]	0.632	CoT	llm	0.681 [0.609, 0.751]	0.666
DIRECT	llm	0.676 [0.601, 0.741]	0.689	DIRECT	llm	0.597 [0.523, 0.669]	0.702
CoT	llm	0.616 [0.552, 0.676]	0.649	FILLER	llm	0.585 [0.513, 0.654]	0.684
TRIVIAL	tabular	0.432 [0.320, 0.548]	0.500	TRIVIAL	tabular	0.399 [0.299, 0.499]	0.500

BRAIN-AGE — brain age, cognitively normal scans (323 scans / 207 subjects)				
Arm	Family	MAE (years) [95% CI]	RMSE	Pearson r
TABPFN v2	tabular	5.28 [4.65, 5.93]	7.01	0.794
XGBOOST	tabular	5.68 [5.04, 6.34]	7.38	0.771
LOGREG	tabular	6.16 [5.51, 6.85]	7.78	0.739
TRIVIAL	tabular	<i>9.07</i> [8.08, 10.09]	<i>11.63</i>	−0.156
CoT	llm	11.43 [10.14, 12.79]	14.61	0.277
FILLER	llm	11.82 [10.15, 13.54]	15.89	0.118
DIRECT	llm	12.08 [10.39, 13.88]	16.31	0.102

Table 2: Out-of-fold accuracy under grouped 5-fold cross-validation, with donor/subject-clustered bootstrap CIs ($B = 2000$). Best per column in **bold**. CoT never reaches the best tabular baseline on any task, and on BRAIN-AGE *every* LLM condition is worse than the trivial predictor that outputs the cohort mean age (italicised).

Task	Contrast	Metric	Difference [95% CI]	p (boot)	q (BH)	Paired test
TAU-STAGE	CoT – DIRECT	Δ AUC	−0.060 [−0.127, +0.009]	0.084	0.151	McNemar $p = 0.056$
TAU-STAGE	CoT – FILLER	Δ AUC	−0.090 [−0.156, −0.024]	0.010	0.048	McNemar $p = 0.015$
TAU-STAGE	DIRECT – FILLER	Δ AUC	−0.030 [−0.057, −0.006]	0.016	0.048	McNemar $p = 0.180$
PROGRESSION	CoT – DIRECT	Δ AUC	+0.084 [+0.008, +0.158]	0.029	0.065	McNemar $p = 0.615$
PROGRESSION	CoT – FILLER	Δ AUC	+0.096 [+0.023, +0.173]	0.014	0.048	McNemar $p = 1.000$
PROGRESSION	DIRECT – FILLER	Δ AUC	+0.012 [−0.045, +0.075]	0.672	0.672	McNemar $p = 0.332$
BRAIN-AGE	CoT – DIRECT	Δ MAE	−0.65 y [−2.18, +0.72]	0.391	0.503	Wilcoxon $p = 0.833$
BRAIN-AGE	CoT – FILLER	Δ MAE	−0.39 y [−1.82, +0.93]	0.586	0.659	Wilcoxon $p = 0.784$

Table 3: Paired condition contrasts on identical rows, clustered paired bootstrap with Benjamini–Hochberg FDR within family. On TAU-STAGE, CoT is significantly worse than semantically empty filler tokens of the same length. On PROGRESSION the sign reverses, but table 4 shows that advantage is largely a confidence-verbalisation effect. On BRAIN-AGE no contrast is significant. Negative Δ MAE favours CoT.

PROGRESSION is similar (agreement 0.757, AOC = 0.090, mistake sensitivity 0.022). On these tasks the chain is post-hoc narration of an answer that already exists.

Paraphrase sensitivity is uniformly low (1.4–5.6%), which rules out the alternative explanation that the chains carry information in an encoded or steganographic form. The information is where it appears to be; there is simply very little of it.

Regression inverts the picture. On BRAIN-AGE, AOC is 0.378 [0.335, 0.425]— $5.7\times$ higher than TAU-STAGE—and mistake injection moves the prediction by 0.33 SD. The chain is genuinely load-bearing when the target is continuous. To our knowledge this is the first evidence that CoT faithfulness differs systematically between classification and regression, and it runs opposite to the accuracy numbers, since BRAIN-AGE is the task the LLM performs worst on. The plausible mechanism is that a two-way choice can be produced in a single forward pass and then narrated, whereas producing a specific number appears to require the intermediate arithmetic-like steps to actually be written.

4.3 CFIT and ESA

table 6 reports the two new metrics. CFIT is positive: perturbing the features a chain names as decisive shifts its prediction $1.39\times$ (TAU-STAGE), $2.13\times$ (PROGRESSION, $p = 0.0005$) and $1.54\times$ (BRAIN-AGE, $p = 0.007$) more than perturbing an equal-sized random set of features the chain

Task	Condition	Distinct conf. values	Conf. entropy	Balanced acc.	ROC-AUC	ECE
PROGRESSION	CoT	10	1.913	0.666	0.681	0.083
PROGRESSION	DIRECT	2	0.644	0.702	0.597	0.314
PROGRESSION	FILLER	8	1.274	0.684	0.585	0.287
TAU-STAGE	CoT	10	1.533	0.649	0.616	0.289
TAU-STAGE	DIRECT	8	1.189	0.689	0.676	0.305
TAU-STAGE	FILLER	8	1.279	0.706	0.706	0.294

Table 4: A confound we checked rather than assumed. AUC is computed from the model’s *stated* confidence, so confidence granularity can drive it. On PROGRESSION the DIRECT condition emits only two distinct confidence values, which mechanically depresses its AUC; on the threshold-free metric DIRECT is in fact better than CoT (0.702 vs. 0.666). What CoT does genuinely deliver on PROGRESSION is far better calibration (ECE 0.083 vs. 0.314)—a real benefit, but a different one from the hypothesised gain in discrimination.

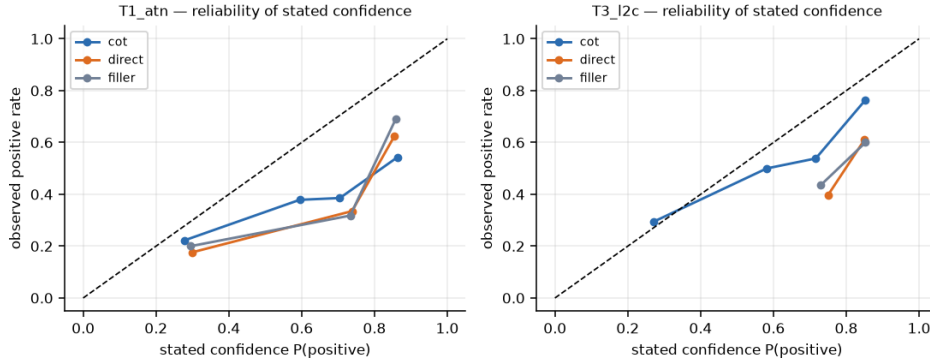


Figure 2: Confidence resolution and calibration. CoT spreads its confidence over ten distinct values on both classification tasks, whereas DIRECT collapses to two on PROGRESSION. The reliability curves show CoT is the only well-calibrated condition on PROGRESSION (ECE 0.083), and that all three conditions are similarly overconfident on TAU-STAGE.

ignores. This is the one clear positive for CoT-as-interpretability: even when the global chain is post-hoc, the specific measurements it points at do carry causal weight. Because our citation parser is a deliberately conservative lexical matcher, paraphrastic citations are undercounted and this estimate is biased toward the null.

ESA is negative on the primary task. Citation frequency and permutation importance are essentially uncorrelated on TAU-STAGE ($\rho = 0.126$, $p = 0.629$) and BRAIN-AGE ($\rho = 0.105$, $p = 0.895$), and only modestly aligned on PROGRESSION ($\rho = 0.491$, $p = 0.039$). The reason is visible in figure 4b: the chains cite nearly every biomarker in nearly every case—total tau and pTau at a 100% rate, AT8 at 98.6%—so citation frequency carries almost no information about which biomarker matters. The chains *do* mention AT8, the single decisive feature by permutation importance (ΔAUC 0.110, four times the next feature); they simply mention it no more than they mention uninformative ones. An explanation that names everything explains nothing.

4.4 Robustness: supported in slope, vacuous in level

table 7 and figure 5 report the perturbation sweeps. The CoT arm is essentially flat under missingness on TAU-STAGE (+2.4% from clean to 50% MCAR) and BRAIN-AGE (−4.0%, i.e., slightly better), while XGBOOST loses 20.6% of its AUC on TAU-STAGE and more than doubles its MAE on BRAIN-AGE. This matches the observation of Kearney and Yang [2026] that LLM prompts represent missingness natively and require no imputation.

But flatness is not an advantage when the level is low. On TAU-STAGE, CoT at 50% missingness (0.609) is still below XGBOOST at 50% missingness (0.630). On BRAIN-AGE the crossover is real and quantitative: XGBOOST degrades past the CoT arm somewhere between 20% and 30% missingness and reaches 14.23 y at 50% against CoT’s 10.83 y. TABPFN v2, however, stays better

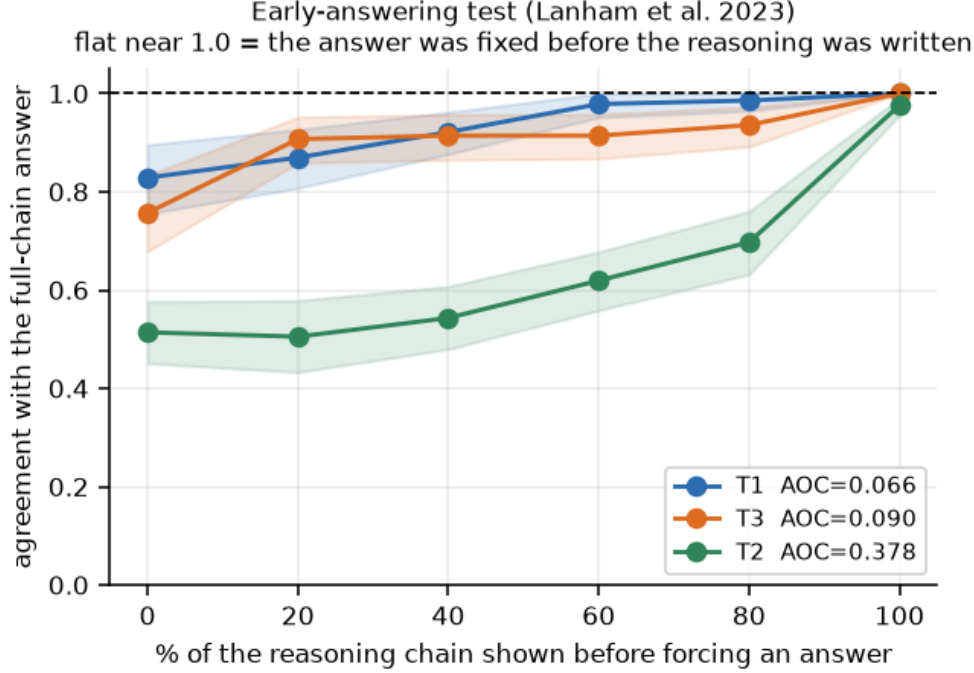


Figure 3: Early answering. Each curve shows how often the answer produced from a truncated chain matches the full-chain answer, as a function of how much of the chain the model is shown. A faithful chain should start low and climb; the two classification tasks start at 0.76–0.83, meaning the answer is essentially fixed before any reasoning is revealed. The continuous target (BRAIN-AGE, using the normalised answer shift of equation 1) is the only task with a substantial area over the curve.

Metric	TAU-STAGE (classif.)	PROGRESSION (classif.)	BRAIN-AGE (regression)
Agreement with full-chain answer at 0% of chain shown	0.829	0.757	0.514
Early-answering AOC [95% CI]	0.066 [0.042, 0.093]	0.090 [0.058, 0.126]	0.378 [0.335, 0.425]
Mistake-injection answer-change rate [95% CI]	0.036 [0.000, 0.079]	0.022 [0.000, 0.056]	0.325 [0.238, 0.428]
Paraphrase answer-change rate [95% CI]	0.014 [0.000, 0.037]	0.021 [0.000, 0.049]	0.056 [0.031, 0.084]

Table 5: Causal faithfulness of the QWEN2.5-7B chains. On the two classification tasks the chains are essentially post-hoc narration: showing the model none of its own reasoning reproduces the full-chain answer 75–83% of the time, and a 10× corruption of a biomarker value inside the chain changes the answer 2–4% of the time. On the continuous target the picture inverts—AOC is 5.7× higher and mistake injection moves the prediction by 0.33 SD. Uniformly low paraphrase sensitivity shows the chains are not using encoded reasoning; the information is where it appears to be, there is simply very little of it.

than the LLM at every level tested (8.01 y at 50% MCAR). The actionable conclusion is therefore about tabular foundation models, not about reasoning. And on PROGRESSION the pattern does not hold at all: CoT is the *least* robust arm (−22.2%), worse than every tabular baseline.

4.5 The clinical endpoint: a brain-age gap with no disease signal

A brain-age model is clinically interesting only if the gap between predicted and chronological age tracks disease. table 8 compares the age-bias-corrected gap between out-of-fold controls and demented scans. The conventional models reproduce the textbook finding of roughly four years of apparent brain ageing in dementia (+4.26 y for TABPFN v2, Cohen’s $d = 0.730$, AUC 0.701). The CoT-derived gap is +0.15 y, with AUC 0.490 [0.435, 0.547] and $p = 0.670$: no disease signal whatsoever. This is the most clinically legible result in the study. The reasoning chains are articulate about atrophy and ageing, and the quantity they produce is diagnostically empty.

Task	Cited	Uncited	Ratio	p (boot)
TAU-STAGE	0.279	0.200	1.39×	0.090
PROGRESSION	0.364	0.171	2.13×	0.0005
BRAIN-AGE	0.669	0.434	1.54×	0.007

(a) CFIT: normalised prediction shift when the features a chain names as decisive are perturbed by ± 1 SD, versus an equal-sized random set it ignores.

Task	Spearman ρ	p	Highest-signal feature
TAU-STAGE	0.126	0.629	AT8 (Δ AUC 0.110)
PROGRESSION	0.491	0.039	lowest-ever MMSE (0.126)
BRAIN-AGE	0.105	0.895	nWBV (6.39 y)

(b) ESA: rank correlation between per-feature citation frequency and permutation importance measured independently on the black-box arm.

Table 6: The two new interpretability metrics. CFIT is the one clear positive for CoT-as-interpretability: the measurements a chain points at are more influential than the ones it passes over, significantly so on PROGRESSION and BRAIN-AGE. ESA is negative on the primary task—the chains cite total tau and pTau in 100% of cases and AT8 in 98.6%, so citation frequency carries almost no information about which biomarker matters. An explanation that names everything explains nothing.

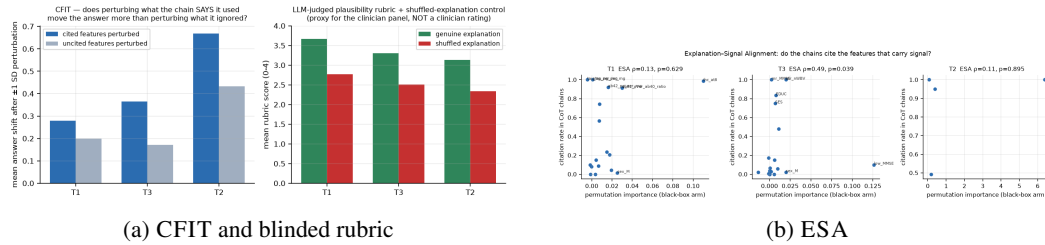


Figure 4: (a) CFIT shift for cited versus uncited features, alongside blinded rubric scores for genuine and shuffled rationales. (b) Per-feature citation frequency against independently measured permutation importance. The chains saturate their citations—most features are named in most cases—so the ranking carries little signal on TAU-STAGE.

4.6 Blinded plausibility: persuasive in proportion to how little it does

table 9 reports the blinded rubric. The shuffled-explanation control passes on all three tasks—genuine rationales score 0.79–0.90 points above rationales written for a different patient, with $p < 10^{-23}$ —so the rubric discriminates rather than rubber-stamping. Given that validity check, the headline is the dissociation. On TAU-STAGE the chains score 3.67/4 overall, including 4.00/4 for absence of fabrication and 3.96/4 for biological validity, while responding to a 10× corruption of their own content 3.6% of the time. The chains are rated as near-perfectly non-fabricating and biologically sound while being almost entirely disconnected from the computation that produced the answer. We note that these are LLM-proxy scores produced by the same base model that generated the chains, which risks self-preference; see section 5.

4.7 Model size: CoT is faithful exactly where it is needed

table 10 and figure 6 report the cleanest causal story in the study, with accuracy and faithfulness reported jointly as Bentham et al. [2024] require. QWEN2.5-1.5B cannot do TAU-STAGE without reasoning: its DIRECT AUC is 0.504, indistinguishable from chance. CoT lifts it to 0.586 (Δ AUC +0.082 [+0.018, +0.143], $p = 0.012$), and its chains are correspondingly load-bearing—AOC 0.246 against the 7B model’s 0.066 (3.7×) and mistake sensitivity 0.218 against 0.036 (6×). QWEN2.5-7B already knows the answer without reasoning (DIRECT AUC 0.676); its chains are decorative, and making it verbalise them costs 0.060 AUC.

CoT is faithful exactly where the model needs it to compute, and decorative exactly where it does not. This is precisely backwards from the deployment incentive: a clinical system would deploy the larger, more accurate model, and would therefore get the least trustworthy explanations.

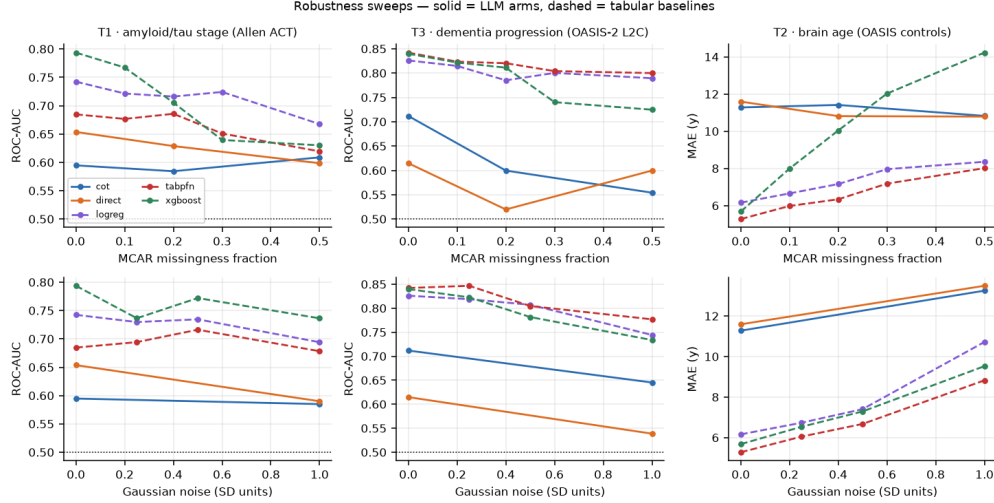


Figure 5: Performance under MCAR missingness and Gaussian feature noise, with every arm receiving the same corrupted values. The LLM curves are nearly horizontal, but they are horizontal at a level the tabular baselines only reach under extreme corruption. TABPFN v2 dominates the LLM arm at every perturbation level on BRAIN-AGE.

Task	Arm	Clean	50% MCAR	Relative change
TAU-STAGE	XGBOOST	0.794	0.630	−20.6%
TAU-STAGE	LOGREG	0.742	0.668	−10.1%
TAU-STAGE	CoT	0.595	0.609	+2.4%
TAU-STAGE	DIRECT	0.654	0.599	−8.4%
PROGRESSION	XGBOOST	0.840	0.725	−13.7%
PROGRESSION	TABPFN v2	0.842	0.800	−5.0%
PROGRESSION	CoT	0.712	0.554	−22.2%
BRAIN-AGE	XGBOOST	5.68 y	14.23 y	+150.5% (worse)
BRAIN-AGE	TABPFN v2	5.28 y	8.01 y	+51.9% (worse)
BRAIN-AGE	CoT	11.28 y	10.83 y	−4.0% (better)

Table 7: Degradation from clean data to 50% MCAR missingness (AUC for TAU-STAGE/PROGRESSION, MAE for BRAIN-AGE). The CoT arm is essentially flat under missingness on TAU-STAGE and BRAIN-AGE—but flat at a level the baselines only fall to under extreme corruption. On PROGRESSION it is the least robust arm of all. The robustness claim is supported in slope and vacuous in level.

4.8 Error analysis

Reasoning past the decisive biomarker (TAU-STAGE). In the first CoT error on TAU-STAGE (truth: early tau; predicted: advanced tau, confidence 85), the chain reads: “The Abeta-42/40 ratio is high (4.575), indicating significant amyloid burden. High total tau (1.67 ng/mg) and phospho-tau (1.689 ng/mg) suggest active tau pathology... The AT8 pTau immunoreactivity is minimal, but the overall tau burden and the presence of amyloid plaques suggest a more advanced stage of neurofibrillary tangles.” AT8 immunoreactivity is the single most informative feature on this task. The chain notices it, explicitly flags it as minimal, and then overrides it with weaker evidence. This is the ESA result made concrete: right facts, wrong weighting.

Plausible, well-hedged and wrong (PROGRESSION). “The recent MMSE score of 27 is within normal limits... nWBV at 0.806 is close to the average... Given the lack of significant changes in MMSE and nWBV, and the absence of other markers of neurodegeneration like amyloid or tau pathology (which are not provided), the risk of dementia is low.” Truth: demented. A systematic

Arm	Family	BAG in demented	BAG in controls	ROC-AUC [95% CI]	Cohen's d	p
TABPFN v2	tabular	+4.26 y	0.00	0.701 [0.646, 0.757]	0.730	$< 10^{-4}$
LOGREG	tabular	+3.90 y	0.00	0.691 [0.633, 0.749]	0.660	$< 10^{-4}$
XGBOOST	tabular	+4.10 y	0.00	0.683 [0.628, 0.737]	0.664	$< 10^{-4}$
CoT	llm	+0.15 y	0.00	0.490 [0.435, 0.547]	0.015	0.670
DIRECT	llm	−0.20 y	0.00	0.470 [0.408, 0.535]	−0.019	0.202

Table 8: The clinical endpoint. Normative models are fitted on cognitively normal scans only and the regression-to-the-mean bias is removed within controls, so the control gap is zero by construction. The conventional models reproduce the textbook finding of roughly four years of apparent brain ageing in dementia; the CoT-derived gap carries no disease signal at all (AUC 0.490, $p = 0.670$). The chains are articulate about atrophy and ageing, and the quantity they produce is diagnostically empty.

Task	Genuine explanation	Shuffled (mismatched)	Difference	Mann–Whitney p	Mistake sensitivity
TAU-STAGE	3.67 / 4	2.78	+0.90	9.7×10^{-24}	0.036
PROGRESSION	3.30 / 4	2.51	+0.79	1.0×10^{-31}	0.022
BRAIN-AGE	3.13 / 4	2.34	+0.79	5.7×10^{-31}	0.325

Table 9: The dissociation. The shuffled-explanation control passes—the rubric reliably separates a genuine rationale from one written for a different patient—so the scores are discriminative rather than a rubber stamp. Given that, the headline is that TAU-STAGE chains score 3.67/4 for plausibility while responding to a $10\times$ corruption of their own content 3.6% of the time. Scores are LLM-proxy ratings, not clinician ratings.

bias accompanies this: on PROGRESSION the CoT arm predicts dementia in only 29.1% of cases against a base rate of 47.1%, i.e., it under-calls progression—the clinically costlier direction of error.

Longer chains do not rescue harder cases. Chain length is negatively associated with correctness: point-biserial $r = -0.114$ ($p = 0.027$) on TAU-STAGE and $r = -0.178$ ($p = 0.008$) on PROGRESSION.

Parsing integrity. Final answer-parse success was 100.0% for QWEN2.5-7B on all three tasks and 99.5% for QWEN2.5-1.5B, so no result depends on discarded generations. Where a generation failed to emit an answer it was re-run with a forced continuation rather than dropped; the fraction requiring that was 0.00%, 0.00% and 0.19% for QWEN2.5-7B on TAU-STAGE/BRAIN-AGE/PROGRESSION, against 45.6% for the QWEN2.5-1.5B CoT condition, which frequently exceeded the 512-token budget.

5 Discussion

5.1 Answering the question

Can chain-of-thought serve as an interpretability layer for Alzheimer’s biomarker models? In this setting, no. CoT did not match black-box accuracy on any of the three tasks, it did not reliably beat its own no-reasoning controls, and on the primary amyloid/tau task its chains failed the causal battery decisively. The combination that matters, though, is not “CoT is inaccurate”—that could be fixed with a better model. It is *high rated plausibility with near-zero causal dependence*: 3.67/4 on a rubric with a validated discriminative control, alongside a 3.6% sensitivity to a tenfold corruption of the chain’s own content. An explanation that is persuasive and inert is worse than no explanation, because it transfers unearned confidence to the clinician.

5.2 What survives

Three components of the hypothesis survive in modified form, and they are why we regard this as a warning rather than a null result.

Robustness is partly vindicated, but the lesson is about tabular foundation models. The LLM arm is genuinely flat under missingness where XGBOOST collapses, and there is a real crossover on brain age beyond roughly 25% missingness. This matches Kearney and Yang [2026]: LLM prompts represent missing cells natively and need no imputation. But TABPFN v2 dominates the LLM at

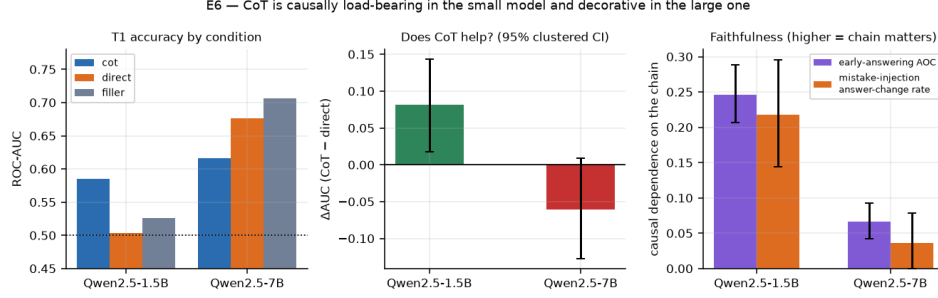


Figure 6: Accuracy benefit of CoT and causal faithfulness of the resulting chains, for QWEN2.5-1.5B and QWEN2.5-7B on TAU-STAGE. Both move together, and both move *down* with capability: the smaller model gains from reasoning and reasons faithfully, the larger model loses from reasoning and narrates.

Quantity (TAU-STAGE)	QWEN2.5-1.5B	QWEN2.5-7B
ROC-AUC, DIRECT	0.504	0.676
ROC-AUC, FILLER	0.526	0.706
ROC-AUC, CoT	0.586	0.616
$\Delta\text{AUC (CoT} - \text{DIRECT)}$	$+0.082 [+0.018, +0.143], p = 0.012$	$-0.060 [-0.127, +0.009], p = 0.084$
Early-answering AOC	0.246 [0.206, 0.289]	0.066 [0.042, 0.093]
Mistake-injection change rate	0.218 [0.144, 0.296]	0.036 [0.000, 0.079]
Paraphrase change rate	0.079	0.014
Median chain length (chars)	1414	642
Forced-answer rate (chain over-ran budget)	0.456	0.000

Table 10: Accuracy and faithfulness reported jointly, as Bentham et al. [2024] require. The 1.5B model cannot do the task without reasoning (DIRECT AUC $0.504 \approx$ chance), CoT lifts it to 0.586, and its chains are $3.7\times$ more causally load-bearing and $6\times$ more sensitive to injected mistakes. The 7B model already knows the answer, its chains are decorative, and verbalising them costs accuracy. The 1.5B forced-answer rate is a caveat on the magnitude, not the direction (section 5).

every level tested, and on PROGRESSION the CoT arm is the least robust of all. “Use an LLM for robustness” is not the right conclusion; “use a tabular foundation model” is.

CFIT supports a weaker, more interesting claim. Even when the chain as a whole is post-hoc, the individual measurements it names are $1.4\text{--}2.1\times$ more influential than the ones it ignores. Local citations are informative even when the global narrative is not. This suggests a constrained citation format—“name the three measurements you used” plus the answer—might retain most of the interpretability value at a fraction of the accuracy cost of free-form narration.

Calibration improves under CoT on PROGRESSION (ECE 0.083 vs. 0.314). Communicating uncertainty well is a genuine clinical good, and it is a benefit the interpretability hypothesis did not anticipate.

5.3 The finding we did not anticipate

CoT faithfulness is markedly higher for regression than for classification (AOC 0.378 vs. 0.066; mistake sensitivity 0.325 vs. 0.036). The entire faithfulness literature is built on multiple-choice benchmarks, and we had to invent a continuous-target version of the battery to ask the question at all. If this replicates, it implies that the field’s pessimism about CoT faithfulness may be partly an artefact of its answer format—and that CoT’s interpretability value may be highest precisely on the continuous clinical endpoints (biomarker levels, atrophy rates, time-to-conversion) that matter most in AD. It is also the finding we consider most likely to replicate and most worth testing first.

5.4 Relation to prior work

We corroborate Anonymous [2026], who found text rationales gave inferior dementia-classification accuracy versus LLM-free baselines, and extend that from speech to quantitative biomarker tables. We corroborate and sharpen the size-versus-faithfulness result of Lanham et al. [2023], now with the accuracy control Bentham et al. [2024] demanded and in a clinical domain: at 1.5B, CoT is

both useful and faithful; at 7B it is neither. And we supply the measurement that Ding et al. [2025] and Kearney and Yang [2026] both omit. Kearney and Yang [2026] disabled reasoning modes; our results suggest that decision was correct for accuracy purposes, and that had they enabled reasoning, the resulting explanations would have looked convincing while being causally inert.

Separately, our amendment A2 is a reportable data-provenance finding: an L2C-style pipeline applied naively to OASIS-2 yields $AUC = 1.000$ for every model, because OASIS derives its dementia label from CDR and the lagged CDR features are the label. Anyone replicating Ding et al. [2025] on OASIS should de-circularise the feature block before reporting numbers.

5.5 Limitations

Data and scope. ADNI, TADPOLE, AIBL and OASIS-3 are all access-gated, so we replicated protocols on open cohorts; absolute numbers are *not* comparable to the published tables of Ding et al. [2025] or Kearney and Yang [2026], and only within-study contrasts are valid. ALLEN ACT is post-mortem tissue rather than in-vivo PET, so TAU-STAGE tests neuropathological rather than imaging A/T/N, and its age variable is censored above 89. The cohorts are small (107 donors, 150 subjects, 207 subjects), so CIs are wide and a ΔAUC of 0.03 is not resolvable here—which is why equivalence margins were pre-declared rather than chosen post hoc. We had no plasma biomarkers, no PET and no multi-omics. We tested one model family at two sizes; a frontier reasoning model might behave differently, and no API keys were available in our environment, so this is the single most important follow-up.

Methodology. The clinician panel could not be convened and was not simulated. The rubric is scored by an LLM, and by the same base model that produced the chains, which risks self-preference; the shuffled control establishes that the rubric discriminates, not that its absolute scores match expert judgement. CFIT and ESA depend on a hand-written lexical citation table that is deliberately conservative, so both are biased toward the null. The QWEN2.5-1.5B COT arm has a 45.6% forced-answer rate, so its faithfulness numbers are computed on chains that were sometimes truncated before the forced answer; three independent metrics support the direction of the model-size conclusion, but its magnitude is uncertain. The LLM robustness sweep used a 150-case subset and fewer perturbation levels than the tabular sweep, so the LLM noise slopes are two-point fits and should be read as coarse. The brain-age bias correction is fitted on control out-of-fold predictions and the controls are then one of the two compared groups—standard practice, but mildly circular, which makes the demented-group gap (+4.26 y for TABPFN v2) the more robust statistic than the AUC. Two targets or feature sets were amended after inspecting baseline behaviour; both changes were made before any LLM was run and both are audited in the results, but they are researcher degrees of freedom and we disclose them as such. Finally, greedy decoding with a single run removes sampling as a confound between arms but gives no seed-to-seed variance estimate; our uncertainty is quantified over cases, not over generations.

What would invalidate these conclusions. A frontier reasoning model showing high faithfulness *and* competitive accuracy on the same three tasks would overturn the central claim; our result may be specific to a 7B model that was not reasoning-tuned.

5.6 Implications for clinical deployment

Any deployment of CoT as a clinical interpretability layer should be required to report a faithfulness measurement alongside its accuracy. The battery used here costs roughly six extra generations per case and is fully automatable, so there is no practical excuse for omitting it. On this evidence, post-hoc SHAP over a tabular foundation model remains both the more accurate and the more honest option. The broader point generalises past this domain: if a chain can be simultaneously 3.67/4 plausible and 3.6% causal, then human-rated interpretability and machine faithfulness are measuring different things, and clinical AI standards that rely on the former are measuring the wrong one.

6 Conclusion

We asked whether chain-of-thought reasoning can serve as an interpretability layer for Alzheimer’s biomarker prediction, and we measured the answer rather than asserting it. Across three tasks built from open cohorts, CoT cost accuracy relative to both black-box tabular baselines ($\Delta AUC -0.16$ to

−0.18; brain-age MAE 11.4 y vs. 5.3 y) and its own compute-matched no-reasoning controls, and on the primary amyloid/tau task its chains were demonstrably post-hoc: the answer was already fixed before the reasoning was written (AOC = 0.066, 82.9% agreement with zero chain shown), and corrupting a biomarker value inside the chain barely moved it (3.6%). A blinded rubric with a validated shuffled-explanation control nevertheless rated those chains 3.67/4. That dissociation is the paper’s central result, and it makes this a warning rather than a null result: CoT here produces explanations that would earn clinician trust the model has not justified.

Three qualifications make the picture more than negative. CoT is markedly more faithful on a continuous target than on classification (AOC 0.378 vs. 0.066), which the multiple-choice-based faithfulness literature could not have detected. The specific measurements a chain cites do causally influence its answer (CFIT 1.4–2.1 $\times$), even when the surrounding narrative does not. And both CoT’s usefulness and its faithfulness decrease with model capability—it helps and is load-bearing at 1.5B, and hurts and is decorative at 7B—which inverts the deployment incentive.

The takeaway for practitioners is one sentence: report faithfulness alongside accuracy, or do not call a reasoning chain an explanation. Four follow-ups are immediate. First, run frontier reasoning models on the identical harness; this is a drop-in change and the single highest-information experiment. Second, test the classification-versus-regression faithfulness gap directly by framing the same underlying quantity as both a binary status and a continuous level. Third, evaluate constrained-citation prompting—“name the three measurements you used”—which CFIT suggests may retain most of the interpretability value without the accuracy cost of free-form narration. Fourth, build the hybrid: a tabular foundation model for the prediction and an LLM for the narration, conditioned on the model’s attributions and scored by this battery, so the explanation is causally anchored by construction.

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A Additional Experimental Detail

Software and hardware. Python 3.12.8, torch 2.13.0+cu130, transformers 5.15.1, scikit-learn 1.9.0, xgboost 3.4.1, tabpfn 8.4.0, numpy 2.5.2, pandas 3.0.5. Two NVIDIA RTX A6000 (48 GB) were used concurrently, one task per GPU, for approximately two hours of end-to-end GPU time. All LLM weights were staged locally; no external API was called at any point.

Generation budget. Approximately 14,500 generations in total: 7,500 in the accuracy arm (5,469 for QWEN2.5-7B across the three tasks plus 2,031 for QWEN2.5-1.5B), 5,353 for the faithfulness interventions, 720 for the blinded judge, and 972 for the brain-age-gap extension. The full set of raw generations—prompt, chain, parsed answer—is retained alongside the per-case intervention outcomes and the judge scores including the shuffled control.

Filler-token lengths. The FILLER condition prefills the assistant turn with a run of “.” tokens whose count equals the measured median CoT length for that task: 159 tokens for TAU-STAGE, 161 for BRAIN-AGE, and 150 for PROGRESSION. This matches the number of extra forward passes to the CoT condition while carrying no semantic content.

Per-dimension rubric scores on TAU-STAGE (genuine rationales). NO_FABRICATION 4.00, BIOLOGY 3.96, COHERENCE 3.70, GROUNDING 3.64, CLINICAL_USE 3.44, SPECIFICITY 3.29. The lowest dimension is specificity, which is consistent with the ESA result: the chains name many measurements without committing to which one decides the case.

Amendment A1 audit. With the original NIA-Reagan target, the best tabular model reaches AUC 0.674 and the trivial baseline is uninformative, confirming that the 75.7% positive rate left too little signal for a donor-clustered bootstrap to resolve differences at the pre-declared $\Delta\text{AUC} = 0.05$ margin. The Braak $\geq$ IV target used in the main text is 45.8% positive at the donor level and yields a well-powered comparison.

Amendment A2 audit. With the full L2C feature block on OASIS-2, logistic regression, XG-BOOST and TABPFN v2 all reach AUC = 1.000, and permutation importance attributes ΔAUC 0.235 to prior-visit dementia status with every other feature at $\approx 10^{-17}$. After removing all CDR- and dementia-derived history features, 18 cognitive, structural, demographic and temporal features remain and XGBOOST reaches AUC 0.840.